AI for MD-PhD Students: An Exposure Curriculum in 9 Sessions

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At a glance

Nine 90-minute sessions show first-semester MD-PhD students the main families of things AI can do in research and medicine, let them try each one, and train them to question every tool they meet. Each session covers one family of AI use. Every session also carries the same two threads: critical thinking and confidentiality. The course ends with a showcase, where each student presents one AI workflow they have adopted and the failures they found in it.

Session format. About 20 minutes of concepts and live demo, 40 minutes hands-on on the student's own work where possible, 20 minutes on that session's critical-thinking challenge, and 10 minutes of share-out and a failure-log entry. Topics already covered on the AI day appear as short recaps marked "Recap of the AI day".

#	Session	What students try	Critical-thinking challenge
1	The landscape	A map of AI families, open and closed models, AI or a traditional tool	Audit AI headlines: what evidence would you need?
2	Prompting and custom assistants	Advanced prompting; an assistant for their own work	Reworded questions and pushing back on a correct answer
3	Finding and reading evidence	Deep research tools, NotebookLM	Citation audit; what did the report leave out?
4	Code and data analysis	Analysis of a public clinical dataset by chat	Find a planted error, confirm the result a second way
5	Agents and automation	An agent completing a multi-step task	Mark each unchecked step; a hidden instruction on a web page

#	Session	What students try	Critical-thinking challenge
6	AI in biology research	Structure prediction, biology models on Hugging Face	Model-card audit; where the model's own confidence says "don't trust me"
7	AI in the clinic	Imaging AI, ambient scribes, decision support	Read a validation study; swap test for bias
8	Writing, figures and presentations	Writing help, Gamma, generated figures	Audit the deck and a generated figure; write the disclosure
9	Showcase	Each student demos one adopted workflow	Present one failure; ask a peer one critical question

Critical thinking in every session

Critical thinking is practised, not lectured: every session ends with a challenge built around how that day's tool fails, and every student keeps a log of the failures they find.

The five questions. Students ask these of every tool, in every session, until they are a habit:

1. What was it trained on, or given to read?
2. How would I know if this is wrong?
3. Who or what is missing from its data?
4. What did using it cost: privacy, time, or a skill I did not practise?
5. Would a non-AI method do this better?

The failure log. Each student records one AI failure per session in a shared template: the tool, what they asked, what went wrong, how they caught it, and which of the five questions it answers. By Session 9 each student has eight entries, and the showcase draws on them. Failures found on their own work count for more than failures from the class exercise.

Confidentiality in every session

Each tool family carries its own privacy risk, so the data rule is restated at the start of every session for that day's tool.

Course data rule: no identifiable patient data and no unpublished lab data in any public AI tool. Students check, for each tool, whether they are signed in with a personal or an institutional account, and whether inputs may be used for training.

Two human gates. A person decides what the AI may see before anything is sent (the input gate), and a person with the right expertise approves the output before it is used (the output gate). How strict each gate is depends on the data:

Data	What the AI may see and do	The human's role
Published papers, public guidelines, public datasets	Any tool approved for the course	Spot-checks the output and its sources
De-identified research data	The structure of the data only; the code runs where the data lives	Verifies results a second way before use
Identifiable patient data or unpublished lab data	Nothing in public tools; only an institution-approved environment, after approvals	Approves every input and output; access is logged
Anything that affects a patient decision	Drafts and suggestions only, never actions	A qualified clinician decides and stays accountable

A human check counts only if the reviewer has the time and knowledge to disagree with the tool. The tiers are a teaching proposal; the institution's data protection officer should confirm them.

Session 1: The landscape

Students leave with a map of what AI can do today, the difference between open and closed models, and a way to decide whether AI is the right tool at all.

Time	Activity
0 to 10	The course: nine families of AI use, the two threads, the showcase at the end, the failure log
10 to 20	Recap of the AI day: how a language model produces text (prediction, not lookup); machine learning versus deep learning from the reading
20 to 40	The map: chat assistants, evidence tools, coding and data analysis, agents, scientific models for biology, clinical AI, generated images and media. Open versus closed models: Hugging Face as the main hub for open models, and why an open model that runs inside the institution matters for confidential data
40 to 60	AI or a traditional tool: four tests (is the task fixed or open-ended, can the answer be checked, what does an error cost, how sensitive is the data). Card sort of 12 research and clinical tasks into AI, traditional, or both
60 to 80	**Challenge: the headline audit.** Groups get four claims about medical AI from news stories or abstracts. For each they write the evidence they would need before believing it, using the reading's red flags: accuracy above 99%, no external test set, no comparison with a simple baseline
80 to 90	The data rule and the four tiers; first failure-log entry

Session 2: Prompting and custom assistants

Students leave able to get reliable output from a chatbot and to build a guardrailed assistant for their own work, knowing how easily both can be steered.

Time	Activity
0 to 10	Recap of the AI day: anatomy of a prompt, system prompts, and the custom GPT each group built
10 to 30	Beyond the basics: give context and examples, ask for a plan before the answer, ask for sources and stated uncertainty, ask for a fixed output format, break long tasks into steps
30 to 60	Hands-on, individually this time: each student builds a small assistant for their own work, such as a study tutor for a course topic or a helper for a lab protocol, with at least three guardrails
60 to 80	**Challenge: how steerable is it?** Ask one factual question in three wordings and compare the answers. Then push back on a correct answer ("Are you sure? I read the opposite") and record whether the model gives in
80 to 90	Share-out of the most surprising change of answer; failure-log entry

Session 3: Finding and reading evidence

Students leave able to use AI to search and digest literature, and to check that every claim it hands them actually comes from a source that says so.

Time	Activity
0 to 10	Recap of the AI day: retrieval-grounded tools (Perplexity, OpenEvidence) and how retrieval-augmented generation works
10 to 25	Deep research tools: what they do, how long they take, what their reports look like, and why a long report with many links is not the same as a complete one

Time	Activity
25 to 45	NotebookLM walkthrough (Part 1 of the walkthrough deck). Since July 2026 Google calls it Gemini Notebook; the address is unchanged
45 to 60	Hands-on: each student builds a notebook from 5 to 10 papers on their own research topic, asks three questions and opens every citation
60 to 80	**Challenge: what is missing?** Ask a plain chatbot for five references on the topic and check each in PubMed. Then compare a deep research report with your own PubMed search: which key papers did it leave out, and does any cited source say something other than what the report claims?
80 to 90	Share-out; failure-log entry

Only published papers and public guidelines go into these tools.

Session 4: Code and data analysis with chat

Students leave able to turn a question into working analysis code by directing a chatbot, and to catch it when the code is wrong.

Time	Activity
0 to 15	The working loop in a Jupyter or Colab notebook: describe the data and the goal, ask for a plan before code, run it, paste back errors, ask for an explanation of each line kept
15 to 20	What to share with the chatbot: column names, types and a few example rows, never the full dataset when it is sensitive
20 to 55	Guided lab on a public, de-identified clinical dataset: load, check missing values, filter, summarise by group, two plots, and one statistical test whose choice and assumptions the chatbot must justify

Time	Activity
55 to 75	**Challenge: the planted error.** A notebook that runs cleanly but uses an unpaired test on paired data. Students find it, fix it, and confirm the result a second way: by hand on a subset, with another library, or in a fresh chat
75 to 90	Reproducibility in brief (seed, versions, a prompt log, a clean rerun); explain-back in pairs; failure-log entry

This merges the two coding sessions of the earlier plan. The filtering, EDA and plotting skills come from the data science course.

Session 5: Agents and automation

Students leave knowing what an AI agent is, having watched one complete a multi-step task, and knowing why agents need tighter limits than a chat window.

Time	Activity
0 to 20	What an agent is: a model plus tools plus a loop that decides the next step. Examples: agentic coding tools that write and run code, browser agents that click through websites, workflow automation tools, and connectors that let a model query databases directly (MCP, shown in the AI day lecture)
20 to 50	Demo and hands-on: an agentic coding tool builds a small analysis project from a one-page specification the students write first. Students compare the result with what they asked for
50 to 70	**Challenge: what did it do without checking?** Students read the agent's step log and mark every action it took without verifying the result. Then a demo of a hidden instruction planted in a web page, and whether the agent follows it

Time	Activity
70 to 85	Limits for agents: least access, approval before any action that changes or sends something, a log of every step, and never on sensitive data
85 to 90	Failure-log entry

Session 6: AI in biology research

Students leave knowing the main kinds of AI models used in biology, able to find and try them on Hugging Face, and able to judge a model from its documentation.

Time	Activity
0 to 10	Recap of the AI day reading: CNNs, transformers, and data leakage through homologous sequences
10 to 30	The tour: protein structure prediction (AlphaFold and its database), protein and DNA language models, single-cell foundation models, and image segmentation for microscopy
30 to 55	Hands-on with Hugging Face: browse the model hub, open a biology model, read its model card, and try a model in a browser demo (a Space) where one exists. Look up a protein of interest in the AlphaFold database
55 to 75	**Challenge: read the model's own warnings.** Model-card audit: training data, intended use, how it was evaluated, stated limitations and licence; what is missing from the card? Then find a region of an AlphaFold structure where the model's own confidence score says not to trust it, and connect it to the AI day question on disordered regions
75 to 90	Was the test set too similar to the training set? Short discussion; failure-log entry

Session 7: AI in the clinic

Students leave knowing the main kinds of clinical AI already in use, how such tools are approved and validated, and how bias and over-trust show up at the bedside.

Time	Activity
0 to 20	The tour: imaging AI, ambient scribes that draft notes from a consultation, decision support and risk scores, and clinician-facing chat tools (recap of OpenEvidence from the AI day)
20 to 35	How clinical AI is approved and validated: regulatory clearance is not proof of patient benefit; internal versus external validation; prospective studies; performance that drops when the hospital or population changes
35 to 55	Hands-on: groups read a published validation study of one clinical AI tool, chosen by the instructor, with a short checklist: where was it tested, on whom, against what reference standard, and did it improve outcomes or only accuracy?
55 to 75	**Challenge: the swap test.** The same clinical vignette given to a chatbot with only age, sex or ethnicity changed. Students record any unjustified difference and ask who is missing from the data
75 to 90	Automation bias and the output gate: who signs, and whether the reviewer has time to disagree; failure-log entry

Session 8: Writing, figures and presentations

Students leave able to use AI for writing and presenting, to audit what it adds, and to disclose its use correctly. They also prepare their showcase deck.

Time	Activity
0 to 15	AI for writing: editing, summarising, translating, drafting an abstract or an email. Disclosure and authorship: what journals, funders and the institution require, starting from the AI usage declaration students already fill in for course notebooks
15 to 45	Gamma walkthrough (Part 2 of the walkthrough deck). Hands-on: each student writes their own outline for the showcase and builds the deck from it
45 to 60	AI-generated scientific figures and schematics: what they can do, and why they often get anatomy, biology or labels wrong
60 to 80	**Challenge: the audit.** Mark every claim, number and image in the Gamma deck that the tool added, then verify it or delete it. Find the errors in a generated biological figure. Write a two-sentence AI disclosure for the showcase
80 to 90	Showcase logistics; failure-log entry

Session 9: Showcase

Each student shows one AI workflow they have adopted for their research, studies or clinical work, how they checked it, and where it failed.

Time	Activity
0 to 5	Welcome and rules
5 to 80	Presentations: 6 minutes each (the workflow, a live demo, how it was verified, one failure from the log, the AI disclosure) plus 2 minutes for one critical question from a peer. This fits about 9 students; a larger class runs in two rooms or presents in pairs

Time	Activity
80 to 90	Instructor feedback on the class's failure logs; course close

Showcase rubric, 5 points per row, 30 in total:

Part	What earns full marks
The workflow	A real, repeated task from the student's own work, with a clear reason to use AI rather than a traditional tool
The demo	Runs live on a realistic input, with no sensitive data
Verification	Says how the output was checked, against what source, and how often
The failure	One concrete failure from the log, how it was caught, and what changed afterwards
Data safety	Names the data tier and both human gates for this workflow
The critical question	The question the student asks a peer is specific and answerable, and draws on the five questions

Assumptions to confirm

- **Class size:** the showcase is timed for about 9 students. More students means two rooms, pairs, or a longer final session.
- **Coding time:** the two coding sessions of the earlier plans are merged into Session 4 to make room for agents. If coding depth matters more, Session 5 can fold into Session 4 and coding gets two sessions again.
- **Tools:** ChatGPT Edu (as on the AI day), NotebookLM (now Gemini Notebook), Gamma, Hugging Face, the AlphaFold database, and one agentic coding tool for Session 5. Check licences and data rules for each before Session 1.
- **Instructor-chosen material:** a public de-identified dataset for Session 4, four headline claims for Session 1, one clinical AI validation study for Session 7, and a generated figure with errors for Session 8.

- **Depth:** most families of AI use are reached at the "tried it with guidance" level. Independent use is expected only for the workflow each student presents at the showcase.
- **Earlier plans:** the two hackathon curricula stay as they are. The NotebookLM and Gamma walkthrough deck is reused in Sessions 3 and 8.